# Práctica tres: Sistema musculoesquelético

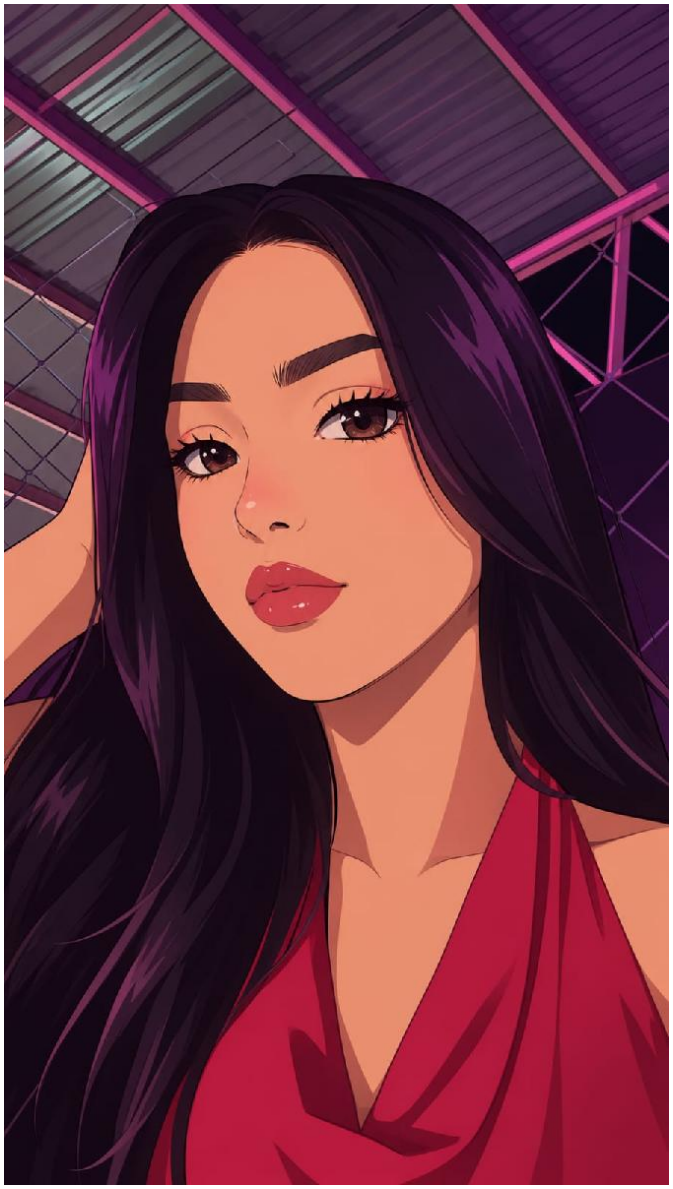
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**Table of Contents**

Información general....1
Datos de la simulación....2
Modelo de EDOS....3
Respuesta a la simulación....4

## Información general



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Asignatura: **Modelado de Sistemas Fisiológicos**

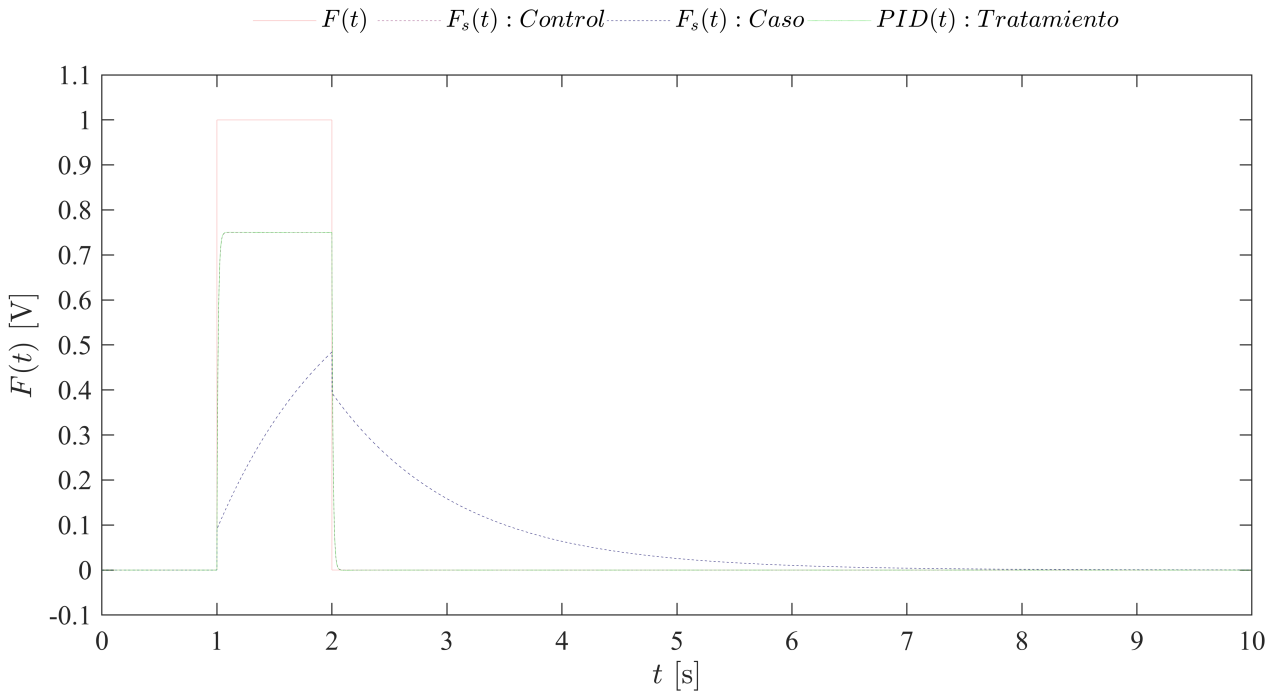
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## Datos de la simulación

```
clc; clear; close all; warning('off','all')
tend = '10';
file = 'Sist3';
open_system(file);
parameters.StopTime = tend;
parameters.Solver = 'ode15s';
parameters.MaxStep = '1e-3';
```

```
x = sim(file,parameters);

X = sim(file, parameters);
plotsignals(X.t, X.F, X.Fs1, X.Fsy, X.PID,'simulink')
```



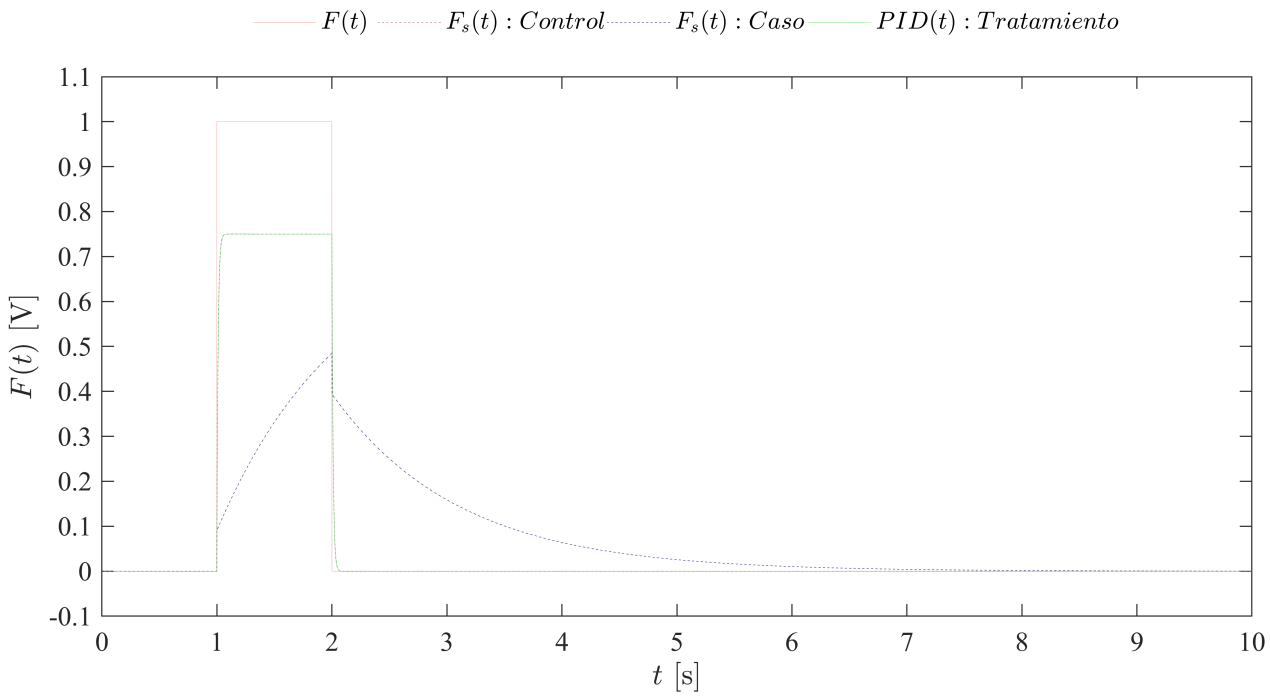

## Modelo de EDOS

```
dt = 1e-3;
t = (0:dt:10)';
F = zeros(length(t), 1); F(1/dt : 2/dt) = 1;
PID = interp1(x.t, x.PID, t, 'previous');

par_control = [0.25, 10e-6, 100e-6, 100];
par_caso = [0.25, 10e-6, 100e-6, 10e3];

Fs_control = musculoesqueletico(par_control, F, t);
Fs_caso = musculoesqueletico(par_caso, F, t);

plotsignals(t, F, Fs_control, Fs_caso, PID, 'EDOsys')
```




```
function Fs = musculoesqueletico(par, F, t)
alpha = par(1); Cs = par(2); Cp = par(3); R = par(4); dt = t(2);
n = round(t(end)/dt);
x = zeros(length(t), 1); x(1) = 0;
Fs = zeros(length(t), 1); Fs(1) = 0;
for i = 1:n
    x(i+1) = x(i) + ((F(i) - x(i)) / (R * (Cp + Cs))) * dt; %EDO
    Fs(i+1) = Cs * F(i) / (Cp + Cs) + (1 - alpha - Cs/(Cp+Cs))*x(i);%Funcion
end
end
```

## Respuesta a la simulación

```
function plotsignals(t, x1, x2, x3, x4, signal)
set(figure(), 'color', 'w')
set(gcf, 'units', 'inches', 'position', [1,1,10,5])
set(gca, 'fontname', 'times new roman', 'fontsize', 12)
hold on; grid off; box on;
clr1 = [248, 178, 178]/255;
clr2 = [175, 113, 157]/255;
clr3 = [64, 61, 136]/255;
clr4 = [38, 212, 38]/255;
order = [clr1; clr2; clr3; clr4];
colororder(order)

plot(t, x1, '-', 'LineWidth', 0.25, 'Color', clr1)
```

```
plot(t, x2, '--', 'LineWidth', 0.25, 'Color', clr2)
plot(t, x3, '--', 'LineWidth', 0.25, 'Color', clr3)
plot(t, x4, ':', 'LineWidth', 0.25, 'Color', clr4)
L = legend('$F(t)$', '$F_s(t):Control$', '$F_s(t):Caso$', '$PID(t):Tratamiento$');
set(L, 'interpreter', 'latex', 'location', 'northoutside', 'orientation',
'horizontal', 'box', 'off')
xlabel('$t$ [s]', 'interpreter', 'latex')
ylabel('$F(t)$ [V]', 'interpreter', 'latex')
xlim([0 10]); xticks(0:1:10)
ylim([-0.1 1.1]); yticks(-0.1:0.1:1.1)

exportgraphics(gcf, [signal, 'matlab.pdf'], 'ContentType', 'vector')
end
```